

Day 4

Data, Types & AI Overview

Measures of Central Tendency

Hours 1-4 · Structured & Unstructured Data · AI/ML/DL · Mean · Median · Mode

Introduction — What is Data?

Data is any raw fact, figure, or information collected from the world around us.

Why Data Matters

Every decision we make is based on data.

Businesses use sales data to plan.

Doctors use patient data to treat.

Without data — we are just guessing.

Data in Daily Life

Your phone tracks your steps → fitness data.

Shopping app shows recommendations → browsing data.

Weather app shows forecast → sensor data.

Social media likes → behavioural data.

Two Big Categories

Data is broadly divided into:

→ Structured Data (organized, table-form)

→ Unstructured Data (free-form, messy)

Both types are equally important.

Structured vs Unstructured Data

Structured Data

Organised in rows and columns (like a table).
Easy to search, sort, and analyse.
Stored in databases and spreadsheets.

Examples:

- Student mark sheet (Name, Marks, Grade)
- Bank transaction records
- Hospital patient records

Think of it as a neat cupboard where everything has a fixed shelf.

Unstructured Data

No fixed format or organisation.
Hard to search directly.
Needs AI to extract meaning from it.

Examples:

- Emails and text messages
- Videos, images, audio files
- Social media posts

Think of it as a pile of papers thrown randomly on the floor.

Quantitative Data — Numbers That Measure

Quantitative data is data that can be measured and expressed as numbers.

Numerical Data — Any data represented by a number

Continuous Variables

Can take ANY value within a range.
Values are measured, not counted.

Examples:

- Your height: 5.7 feet, 5.71, 5.712...
- Body temperature: 37.2°C
- Distance: 10.5 km

Key Rule: Values can be infinitely precise (decimal points possible).

Discrete Variables

Can only take SPECIFIC whole values.
Values are counted, not measured.

Examples:

- Number of students in a class: 30
- Goals scored in a match: 2
- Number of books on a shelf: 15

Key Rule: You cannot have 2.5 children or 3.7 goals — only whole numbers.

AI · Machine Learning · Deep Learning

Understanding the three layers of modern intelligence

Artificial Intelligence

The broadest concept.
Making machines smart enough to perform tasks that normally need human intelligence.

Examples:

- Google Maps suggesting routes
- Spam email filter
- Face unlock on your phone

AI is the parent — ML and DL are its children.

Machine Learning

A subset of AI.
Machines learn from data and improve automatically without being explicitly programmed.

Examples:

- Netflix recommending movies
- Email auto-sort (Social/Promo)
- Predicting house prices

Give it data → it finds patterns
→ makes predictions.

Deep Learning

A subset of ML.
Uses multi-layered neural networks inspired by the human brain.
Handles very complex data.

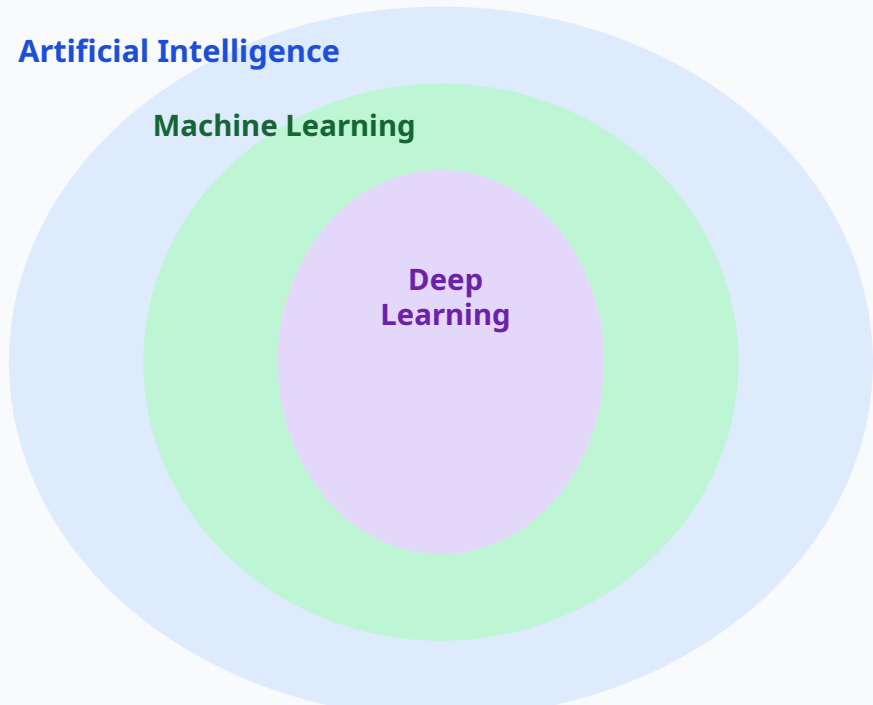
Examples:

- ChatGPT understanding language
- Self-driving cars seeing roads
- Medical image diagnosis

Needs huge amounts of data and powerful computers (GPUs).

How AI, ML & Deep Learning Relate

Think of it as three nesting circles — each one is a more specific version of the one it contains



AI (Artificial Intelligence)

The entire field of making machines intelligent. Broadest category.

Machine Learning

A technique inside AI. Machines learn patterns from data automatically.

Deep Learning

A powerful technique inside ML. Uses brain-inspired layers for complex tasks.

Real-world flow: Collect Data → Choose ML model → Train (Deep Learning if complex) → Predict

Qualitative Data — Categories & Labels

Qualitative data describes qualities or characteristics — it uses words and labels, not numbers.

Categorical

Data placed into groups/categories

Each value belongs to one group.

No numerical meaning.

Example: Blood type (A, B, AB, O)

Example: Favourite colour (Red, Blue)

Nominal

Categories with NO natural order

Groups that are simply different.

Cannot be ranked.

Example: Country names

Example: Car brands (Toyota, Honda)

Ordinal

Categories WITH a natural order

Groups that can be ranked.

But gaps between ranks are unequal.

Example: Education (School→BSc→MSc)

Example: Satisfaction (Low/Med/High)

Binary

Only TWO possible values

The simplest category.

Only two choices exist.

Example: Yes / No

Example: True / False, Pass / Fail

Data Types — Complete Overview

Data Type	Sub-Type	Key Feature	Simple Example
Quantitative	Continuous	Measured — infinite decimal values	Height: 5.7 ft
Quantitative	Discrete	Counted — whole numbers only	Students: 30
Qualitative	Nominal	No order — just different labels	Country: Pakistan
Qualitative	Ordinal	Has a natural rank order	Grade: A > B > C
Qualitative	Binary	Only two possible values	Pass / Fail

Hours 3 – 4

Measures of Central Tendency

Mean · Median · Mode

"A single number that represents the entire dataset"

Mean

Average value

Median

Middle value

Mode

Most frequent

Mean — The Average

The mean is the sum of all values divided by the number of values.

Formula: Mean = Sum of all values ÷ Total number of values

Step-by-Step Example

Marks of 5 students: 60, 70, 80, 90, 100

Step 1 — Add all values:

$$60 + 70 + 80 + 90 + 100 = 400$$

Step 2 — Divide by the count (5):

$$400 \div 5 = 80$$

Mean = 80

Real Life Example

Average monthly salary of a team,
average temperature of a week.

⚠ Common Mistake

Mean is affected by extreme values (outliers). One very high/low number can pull the mean up or down.

Median — The Middle Value

The median is the middle value when data is arranged in ascending (small → big) order.

Rule: Odd count → middle item is median | Even count → average of two middle items

Example — Odd Count (5 values)

Data: 8, 3, 5, 1, 7

Step 1 — Sort ascending:

1, 3, 5, 7, 8

Step 2 — Find middle position:

5 values → position 3

Median = 5

Example — Even Count (6 values)

Data: 4, 8, 2, 6, 10, 1

Step 1 — Sort ascending:

1, 2, 4, 6, 8, 10

Step 2 — Two middles: 4 and 6

$(4 + 6) \div 2 = 5$

Median = 5

Advantage of Median: Unlike the mean, it is NOT affected by extreme (outlier) values.

Mode — The Most Frequent Value

The mode is the value that appears most often in the dataset.

Rule: Find the value that repeats the MOST. A dataset can have one, many, or no mode.

One Mode (Unimodal)

Data:

2,3,4,4,5,6,4,7

Mode = 4 (appears 3 times)

Most common value in dataset.

Two Modes (Bimodal)

Data:

1,2,2,3,4,4,5

Mode = 2 and 4 (each appears twice)

Both values share the top frequency.

No Mode

Data:

10,20,30,40,50

No mode — all values appear once

No repetition means no mode.

Mode is the only measure that works for Qualitative (categorical) data too!

When to Use Mean, Median or Mode?

Measure	What it finds	Best used when...	Daily Example
Mean	The mathematical average	Data has no extreme outliers	Average exam score of the class
Median	The middle value	Data has outliers (skewed)	Median house price in a city
Mode	Most frequently occurring value	Finding the most popular item	Most popular shoe size in a shop

Quick Memory Trick

Mean = Add all then divide | Median = Sort then find middle | Mode = Count and find most repeated

Day 4 — Summary & Key Takeaways

Data Types

Structured data is organised in tables; Unstructured is free-form (videos, texts).

→ → AI DML

AI is the big idea. ML learns from data. Deep Learning handles complex tasks using neural networks.

Mean

Sum ÷ Count. Affected by outliers. Best for evenly spread data.

Quantitative

Numbers — Continuous (measured, decimals OK) vs Discrete (counted, whole numbers).

Qualitative

Labels — Nominal (no order), Ordinal (ranked), Binary (2 values only).

Median

Middle value after sorting. Not affected by outliers. Best for skewed data.

Mode

Most repeated value. Works even for qualitative/categorical data.